

30 Days SQL Micro Course Answer

1. Total Number of Patients (Write an SQL query to find the total number of patients across all hospitals.)

ANSWER:-

```
SELECT SUM (Patients _Count) AS Total _Patients  
FROM Hospital _Data;
```

2. Average Number of Doctors per Hospital (Retrieve the average count of doctors available in each hospital.)

ANSWER:-

```
SELECT Hospital _Name,  
       AVG (Doctors _Count) AS avg _doctors _count  
FROM Hospital _Data  
GROUP BY Hospital _Name;
```

3. Top 3 Departments with the Highest Number of Patients (Find the top 3 hospital departments that have the highest number of patients.)

ANSWER:-

```
SELECT Department, Patients _Count  
FROM Hospital _Data  
ORDER BY Patients _Count DESC  
LIMIT 3;
```

4. Hospital with the Maximum Medical Expenses (Identify the hospital that recorded the highest medical expenses.)

ANSWER:-

```
SELECT Hospital _Name, Medical _Expenses  
FROM Hospital _Data  
ORDER BY Medical _Expenses DESC  
LIMIT 1;
```

5. Daily Average Medical Expenses (Calculate the average medical expenses per day for each hospital.)

ANSWER:-

```
SELECT Hospital _Name,  
        AVG (Medical _Expenses) AS avg _expenses  
FROM Hospital _Data  
GROUP BY Hospital _Name, Admission _Date;
```

6. Longest Hospital Stay (Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.)

ANSWER:-

```
SELECT Hospital _Name, Location,  
        (Discharge _Date-Admission _Date) AS longest _stay  
FROM Hospital _Data  
ORDER BY longest _stay DESC  
LIMIT 1;
```

7. Total Patients Treated Per City (Count the total number of patients treated in each city.)

ANSWER:-

```
SELECT Location, SUM (Patients _Count) AS patients _treated  
FROM Hospital _Data  
GROUP BY Location;
```

8. Average Length of Stay Per Department (Calculate the average number of days patients spend in each department.)

ANSWER:-

```
SELECT Department,  
        AVG (Discharge _Date-Admission _Date) AS AVG_STAY_DEPT  
FROM Hospital _Data  
GROUP BY Department;
```

9. Identify the Department with the Lowest Number of Patients (Find the department with the least number of patients.)

ANSWER:-

```
SELECT Department, SUM (Patients _Count) AS Lowest _number  
FROM Hospital _Data
```

GROUP BY Department

ORDER BY Lowest _number ASC LIMIT 1;

10. Monthly Medical Expenses Report (Group the data by month and calculate the total medical expenses for each month.)

ANSWER:-

SELECT TO_CHAR (Admission _ Date, 'YYYY-MM') AS Months,

SUM (Medical _Expenses) AS total _expenses

FROM Hospital _Data

GROUP BY Months

ORDER BY Months;